Shreya Rose Jimson

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Objective

Final year B.Tech ECE student with a background in semiconductor device modelling and characterization, seeking a 6 month internship (January–June 2027).

Education

PES University — **B.Tech - Electronics and Communication Engineering, CGPA: 8.54/10**
Bengaluru, Karnataka, India — *August 2023 – June 2027*

- Relevant Courses: Semiconductor Devices and Fabrication, Analog Circuit Design, Electronic Principles And Devices, Low Power VLSI, Transmission Lines, Waveguides and Antennas, Electromagnetic Field Theory, Machine Learning and Applications, and Artificial Neural Networks

CMR National PU College — **Class 12, Karnataka Pre University Board, Score: 95.2%**
Bengaluru, Karnataka, India — *June 2021 – May 2023*

Gopalan National School — **Class 10, Indian Certificate of Secondary Education (ICSE), Score: 96.7%**
Bengaluru, Karnataka, India — *June 2008 – May 2021*

Honours and Awards

- PES Prof. CNR Rao Scholarship (Semester V, VI - Merit Scholarship for top performers of the semester)
- PES Distinction Award Certificate (DAC) Scholarship (Semester I, II, III, IV)

Projects

Low Power Hybrid 10T CMOS Full Adder in 45nm Technology

- Implemented a published hybrid 10T full adder in 45nm (Cadence Virtuoso) and optimized it with supply voltage scaling and HVT transistors to reduce dynamic and leakage power
- Verified SUM and CARRY outputs across all input combinations through transient simulation

TCAD Modeling and Characterization of Si/SiGe Line TFET

- Built a Sentaurus TCAD model of a Si/SiGe heterojunction line TFET from a published IEEE design, benchmarked against the paper's reported results
- Resolved simulation convergence issues and analyzed how electric field distribution and band-to-band tunneling control the device's switching behavior

3D TCAD Modeling of Sub-5nm GAA Nanowire FET

- Built a 3D Sentaurus TCAD model of a sub-5nm GAA nanowire FET from a published IEEE design, benchmarked against the paper's reported results
- Extracted threshold voltage, on/off current ratio, and transconductance to study short-channel effects

Semiconductor Manufacturing Failure Prediction using Machine Learning

- Built an SVM model on the SECOM semiconductor manufacturing dataset to predict yield failures, handling a 14:1 pass-to-fail imbalance and comparing PCA with feature selection

Experience

Research Intern – Compact Modeling of 4 - Channel Nanowire FET — May 2026 – August 2026
Chips to Startup (C2S), PES University — *Bengaluru, Karnataka, India*

- Calibrated a BSIM-CMG Verilog-A compact model from Sentaurus TCAD device data and compiled it with OpenVAF for circuit simulation in ngspice
- Automated extraction of threshold voltage, subthreshold swing, and saturation current using a factorial sweep and least-squares fitting in Python, matching TCAD I-V within 3% RMS error
- Verified the calibrated model at circuit level through a resistive-load inverter
- Modeled self-heating via electrothermal TCAD, showing a 100 K peak lattice temperature rise and a 2% on-current reduction at VDD = 0.7 V

Research Intern – Vision Transformer Pedestrian Detection June 2025 – August 2025
Crucible Of Research and Innovation (CORI), PES University *Bengaluru, Karnataka, India*

- Implemented a DETR-based pedestrian detection pipeline in PyTorch and debugged the hybrid CNN-Transformer through tensor-level tracing; work published at IEEE CICN 2027

Research Intern – Quadcopter Simulation June 2024 – August 2024
Center for Robotics, Automation and Intelligent Systems (cRAIS) PES University *Bengaluru, Karnataka, India*

- Simulated quadcopter dynamics and autonomous path navigation in MATLAB/Simulink with Simscape, conducting parameter sweeps to evaluate the effects of package mass and target speed on performance

Skills

EDA & Simulation: Synopsys Sentaurus TCAD, Cadence Virtuoso, ngspice, OpenVAF, LTspice, MATLAB/Simulink
Programming and Tools: Python, LaTeX, MATLAB
Core Competencies: Compact Modeling (BSIM-CMG), Analog Circuit Analysis and Testing, Device Characterization, Semiconductor Device Physics
Soft Skills: Teamwork, communication (English, written and verbal), adaptability, leadership, cross-functional collaboration, proactive, curious, organized, detail-oriented

Publications

Design and Evaluation of A Real-Time Pedestrian Detection System Using Vision Transformer
IEEE 17th International Conference on Computational Intelligence and Communication Networks (CICN)

Certifications

- Advanced Workshop on Metrology and Nano-Characterization Techniques - Centre for Nano Science and Engineering (CeNSE), IISc (October 2026)
- Summer School 2026 - Centre for Nano Science and Engineering (CeNSE), IISc (June 2026)
- India Semiconductor Workforce Development Program (ISWDP) Samsung Student Fellowship Grade II recipient - Cohort 6 (August - December 2025)
- Electrical Engineering Summer School - Indian Institute of Science (July 2025)
- Semiconductor Fabrication 101 - Purdue University (February 2026)
- Quantum Technology Workshop - Quantum Research Park (QuRP), IISc (June 2024)

Leadership & Activities

Head of Event Production 2023 – 2024
TEDxPESU *Bengaluru, Karnataka, India*

- Progressed from volunteer to leading a cross-disciplinary student team (engineering, design, psychology, commerce) through registration, budgeted catering, university permissions, and on-day logistics for TEDxPESU 2024

Head of Creatives 2024 – 2025
IEEE Robotics and Automation Society (RAS) Student Branch, PES University *Bengaluru, Karnataka, India*

- Progressed from member to leading the chapter's design and content work

Technical Domain Member 2024 – 2026
Silicon, VLSI Club, PES University *Bengaluru, Karnataka, India*

References

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